

Overview

This mindmap illustrates the key roles and responsibilities in Software Testing based on the ISTQB (International Software Testing Qualifications Board) framework.

Note on QA vs QC: ISTQB Foundation Level (CTFL 4.0) does not define "QA Role" or "QC Role" as formal certification roles. Instead, it defines **Test Management Role** and **Testing Role**. QA (Quality Assurance) and QC (Quality Control) are **quality concepts** — QA focuses on **process & prevention**, while QC focuses on **product & detection**. They are complementary and are embedded within the roles below, not standalone ISTQB roles.

Mindmap



Role Summary Table

Role	Focus	Key Responsibilities	ISTQB Level
Test Manager	Strategy & Leadership	Test planning, resource management, stakeholder management	Advanced
Test Analyst	Design & Analysis	Requirements analysis, test case design, traceability	Foundation / Advanced
Test Automation Engineer	Tool & Automation	Framework development, script creation, CI/CD	Specialist
Test Environment Manager	Infrastructure	Environment setup, data management, monitoring	Support Role

Role	Focus	Key Responsibilities	ISTQB Level
Technical Test Analyst	Architecture & Non-Functional	Test architecture, performance, security, automation strategy	Advanced

QA vs QC Comparison

Aspect	QA (Quality Assurance)	QC (Quality Control)
Focus	Process & Prevention	Product & Detection
Goal	Prevent defects	Find and report defects
Approach	Proactive	Reactive
Activities	Process improvement, standards	Testing, inspection, review
When	Throughout lifecycle	During execution phase
Output	Process documentation, metrics	Test reports, defect logs

ISTQB Certification Pathway



Key Takeaways

- **ISTQB defines two primary roles:** Test Management Role and Testing Role — not "QA" or "QC" as standalone roles
- **QA** (process & prevention) and **QC** (product & detection) are quality concepts embedded within testing roles, not separate ISTQB-certified roles
- **Verification** ("building the product right") applies to unit, integration, and system testing
- **Validation** ("building the right product") is the primary goal of UAT and acceptance testing
- ISTQB provides a **structured certification pathway** from Foundation to Expert level

- The **Test Manager** oversees testing activities; the **Test Analyst** focuses on test analysis and design
- **Test Automation** is a specialized role within the testing framework